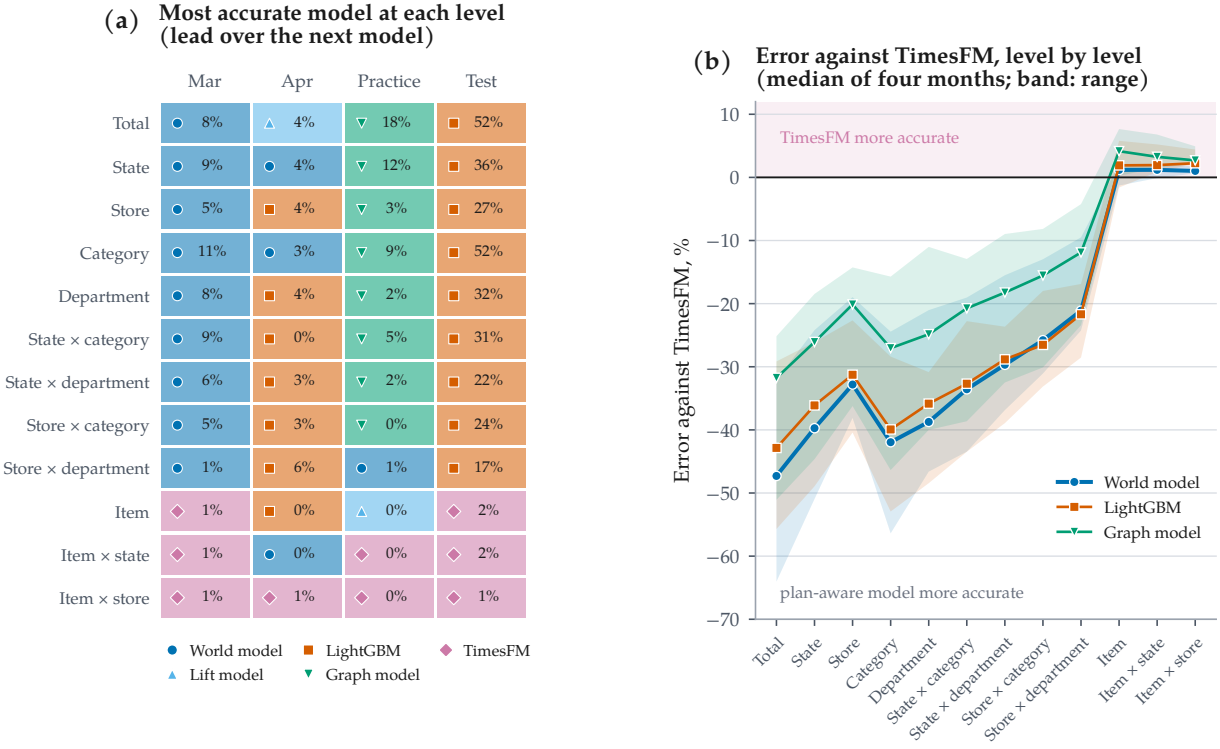
(a)
Most accurate model at each level
(lead over the next model)
Mar
Apr
Practice
Test
Total
State
Store
Category
Department
State × category
State × department
Store × category
Store × department
Item
Item × state
Item × store
8%
4%
18%
52%
9%
4%
12%
36%
5%
4%
3%
27%
11%
3%
9%
52%
8%
4%
2%
32%
9%
0%
5%
31%
6%
3%
2%
22%
5%
3%
0%
24%
1%
6%
1%
17%
1%
0%
0%
2%
1%
0%
0%
2%
1%
1%
0%
1%
World model
LightGBM
TimesFM
Lift model
Graph model
(b)
Error against TimesFM, level by level
(median of four months; band: range)
TimesFM more accurate
plan-aware model more accurate
Error against TimesFM, %
10
0
−10
−20
−30
−40
−50
−60
−70
World model
LightGBM
Graph model
Total
State
Store
Category
Department
State × category
State × department
Store × category
Store × department
Item
Item × state
Item × store